

(TR-104) PYTHON WITH AI/ML

Training Day 22 Report

Date: 10 February 2026

Topic Covered: NLP Foundations – Vectorization, BoW, TF-IDF & Similarity

The twenty-second day of training focused on strengthening core Natural Language Processing concepts, especially how text is converted into numerical form and how similarity between texts is calculated.

- Understood why vectorization is required in NLP (text to numbers conversion).
- Learned the concept of vector representation of documents using features.
- Studied Bag of Words (BoW) approach including vocabulary creation and word frequency representation.
- Explored the vocabulary explosion problem in large datasets.
- Learned TF (Term Frequency) and IDF (Inverse Document Frequency) concepts in detail.
- Understood why TF-IDF is more effective than simple BoW representation.
- Studied cosine similarity and its importance in measuring text similarity.
- Compared similarity vs distance (Euclidean, Manhattan, Cosine).
- Learned why cosine similarity is preferred for text-based data.
- Practiced BoW and TF-IDF concepts for better conceptual clarity.
- Understood how text ranking works based on similarity scores.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/NLP/tree/main/Vectorization>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/introduction-to-natural-language-processing-teaching-machines-to-read-talk-dec3b9ea9453>